De La Salle Holy Cross College

Grade 10 Mathematics November Examination – Paper 1

Examiner: Mathematics Department Total: 100

Moderators: Grade 10 Maths Teachers Time: 2 hours

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Instructions and Information:

- This assessment consists of 12 pages which include 6 questions.
- Answer every question in the spaces provided on this question paper.
- You may use any permitted calculator. Ensure that it is set in degree mode.
- All necessary working-out must be shown clearly; providing only final answers does not necessarily guarantee full marks.
- Diagrams and graphs are not necessarily drawn to scale.
- If necessary, round off answers to **ONE** decimal place, unless otherwise stated.
- Write neatly and legibly; anything that is unclear will not be marked.

	Q 1	Q 2	Q 3	Q 4	Q 5	Q6	Total
Mark							
Total	30	9	14	26	14	7	100
Marker	KR	AB	LC	AB	LC	LC	

QUESTION 1 [30 Marks]

1.1 Simplify the following as fully as possible:

a) $(2x-3y)(x+2y)$ (3)

$= 2x^2 + 4xy - 3xy - 6y^2$

$= 2x^2 + xy - 6y^2$ ✓ ✓ ✓

b) $\frac{3^{x-1}.2^{x+1}}{6^x}$ (3)

$= \frac{3^x . 3^{-1} . 2^x . 2^1}{2^x . 3^x}$ ✓ factors

$= \frac{2}{3}$ ✓ ans

c) $(a^{-1}+b^{-1})^{-1}$ (3)

$= \left(\frac{1}{a} + \frac{1}{b}\right)^{-1}$

$= \left(\frac{b+a}{ab}\right)^{-1}$ ✓ num ✓ den

$= \frac{ab}{a+b}$ ✓ ans

d) $\frac{121^x-1}{11^x-1}$ (2)

$= \frac{(11^x-1)(11^x+1)}{11^x+1}$ ✓ fact

$= 11^x+1$ ✓ ans

1.2 Factorise as fully as possible:

a) $5x - 5y - ty + tx$ (3)

$= 5(x-y) + t(x-y)$ ✓ ✓ fact

$= (x-y)(5+t)$ ✓ ans

b) $16x^2 - (x-3)^2$ (4)

$= [4x + (x-3)][4x - (x-3)]$ ✓ ✓ fact

$= (4x + x - 3)(4x - x + 3)$

$= (5x-3)(3x+3)$ ✓ simp

$= 3(5x-3)(x+1)$ ✓ fact

1.3 Solve the following equations, stating any necessary restrictions:

a) $x(x-2) = 35$ (3)

$x^2 - 2x - 35 = 0$ ✓ reorder

$(x-7)(x+5) = 0$ ✓ fact

$x = 7 ; -5$ ✓ CA ans

b) $\frac{1}{x+2} + \frac{3}{x^2-4} = \frac{2}{2-x}$ (5)

$\frac{1}{x+2} + \frac{3}{(x+2)(x-2)}$ ✓ $= \frac{2}{-(x-2)}$ ✓ ✓ change of sign ✓ fact

$x - 2 + 3 = -2(x+2)$ ✓ ✓ ; $x \neq \pm 2$

$x + 1 = -2x - 4$

$3x = -5$

$x = -\frac{5}{3}$ ✓

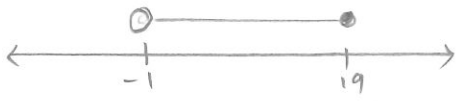
penalise once for no restrictions.

1.4 Solve the following inequality and represent your answer on a number line: (4)

$-15 \leq 4 - x < 5$

$-19 \leq -x < 1$ ✓

$-1 < x \leq 19$ ✓ ans



✓ CA

QUESTION 2 [9 Marks]

2.1 Margot posted a picture of herself on Instagram and each day it gained 15 likes. Hamzah, who follows Margot's Instagram closely, reported that on day one there were 38 likes.

a) How many likes were there on day 2 and day 3? (1)

53 ; 68 ✓ both correct

b) Determine the general term for the number of likes if the pattern continues. Give your answer in its simplest form. (2)

38 ; 53 ; 68

+15 ✓ +15

$T_n = 15n + 23$ ✓

c) On which day will Margot's picture get 518 likes? (2)

$518 = 15n + 23$ ✓m

$33 = n$ ✓ CA

$T_n = a + (n-1)d$

$= 38 + (n-1)15$

$= 38 + 15n - 15 = 15n + 23$

2.2 Consider the number pattern 3; 12; 48; 192; ...

a) What type of sequence is represented above? (1)

exponential / geometric sequence ✓ any

b) Determine the general term (T_n) of this number sequence. (3)

$r = 4$

$T_n = 3 \cdot 4^{n-1}$ ✓ ✓ ✓

QUESTION 3

[14 Marks]

FORMULAE YOU MAY NEED:

$$A = P(1 + in) \qquad A = P(1 + i)^n$$

3.1 Isabella invests R600 at 12% simple interest per year. How much will she have after 3 years? (2)

$A = P(1 + in)$

$= 600(1 + 12\% \times 3)$ ✓ sub

$= R816$ ✓ ans

3.2 At what rate of compound interest will Console's investment of R4000 increase to R6550 over 7 years? (3)

$A = P(1 + i)^n$

$6550 = 4000(1 + i)^7$ ✓ sub

$\sqrt[7]{\frac{6550}{4000}} - 1 = i$

$i = 0,07299...$ ✓

$\therefore r = 7,3\%$ ✓

penalise 1 for dp!

3.3 Laetitia wants to invest some of her allowance money for when she has completed her studies at university. She invests an amount of R8000 at compound interest for 8 years. The interest rate is 7% for the first 4 years, 6% for the next 3 years and 5.6% for the last year. Calculate the amount that Laetitia would have after 8 years. (3)

$A = 8000(1 + 7\%)^4(1 + 6\%)^3(1 + 5,6\%)^1$ ✓ correct sub

$= R13\ 188;84$ ✓ CA

3.4 Jaimie wants to buy a Samsung Galaxy Tablet which is priced at R6999,90. There is an option of paying a 15% deposit and then making 24 monthly payments using a hire purchase agreement where interest is calculated at 7,25% p.a. simple interest.

a) Calculate what Jaimie's monthly payments will be. (5)

Deposit = 15% of R6 999,90
= R1 049,99 ✓

Balance = R6 999,90 - R1049,99
= R5 949,91 ✓

A = R5 949,91 (1 + 7,25% x 2) ✓ sub
= R6 812,65 ✓ CA

Monthly payments = R6 812,65 ÷ 24
= R283,86 ✓

b) Calculate the total amount that Jaimie will pay for the Galaxy Tablet. (1)

Total = R1049,99 + R6812,65
= R7 862,64 ✓ CA

QUESTION 4 [26 Marks]

4.1 Given: $f(x) = \frac{3}{x} + 1$ and $g(x) = -2x - 4$

a) Solve for x if $f(x) = g(x)$. (3)

$\frac{3}{x} + 1 = -2x - 4$ ✓

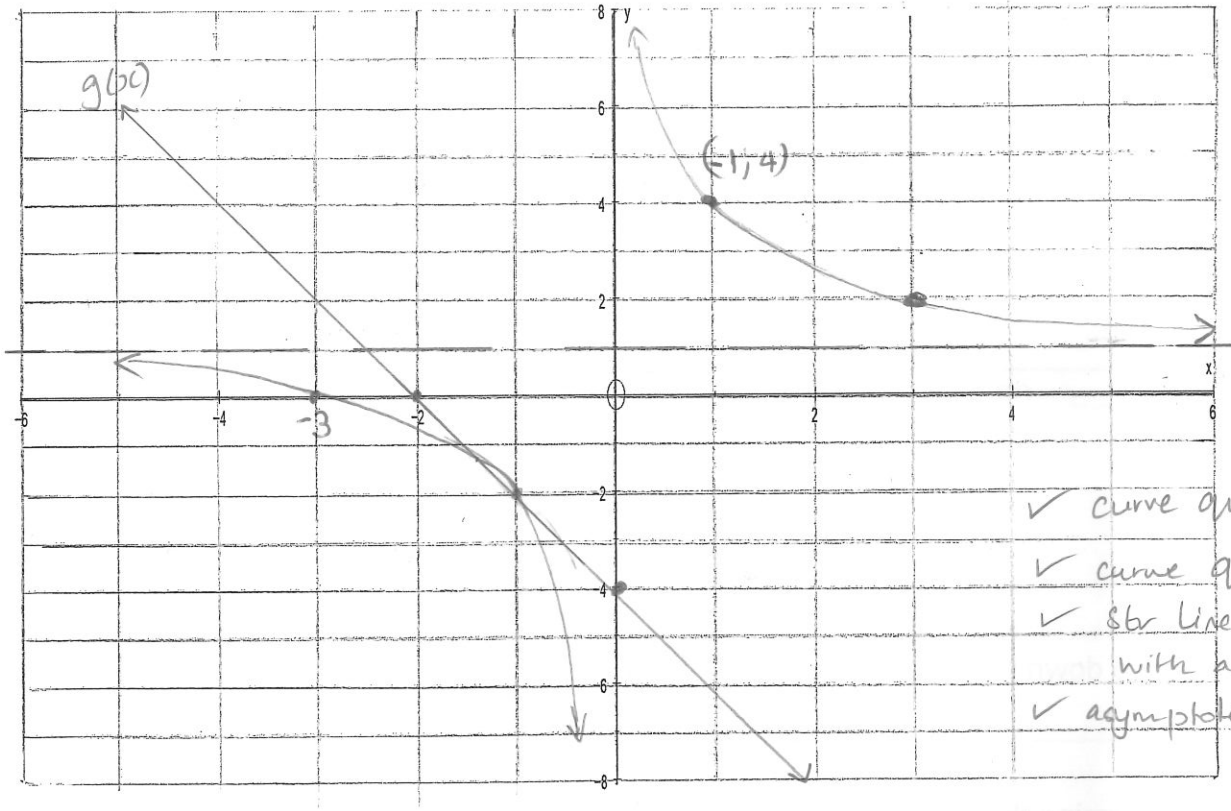
$3 + x = -2x^2 - 4x$

$2x^2 + 5x + 3 = 0$

$(2x + 3)(x + 1) = 0$ ✓

$x = -3/2 \; ; \; -1$ ✓ CA

b) Sketch the graphs of f and g, clearly showing all intercepts and asymptotes. (4)



✓ curve quad 1
✓ curve quad 3
✓ Str Line intercept with axes
✓ asymptote

c) Write down the equations of the asymptotes of f. (2)

$x = 0$ ✓

$y = 1$ ✓

d) Write down the domain of f. (2)

$x \in \mathbb{R} \; ; \; x \neq 0$ ✓ ✓

OR

$x \in (-\infty; 0) \cup (0; \infty)$ or $x \in \mathbb{R} - \{0\}$ or similar

e) Using your graph, determine the values of x for which $-4 \leq g(x) < 0$. (2)

Read off from graph : $-2 < x \leq 0$ ✓ ✓

f) Determine the y intercept of k if $k(x) = 2g(x)$. (1)

-8 ✓

g) Determine the x and y intercepts of h if h is the graph of g reflected about the y-axis. (2)

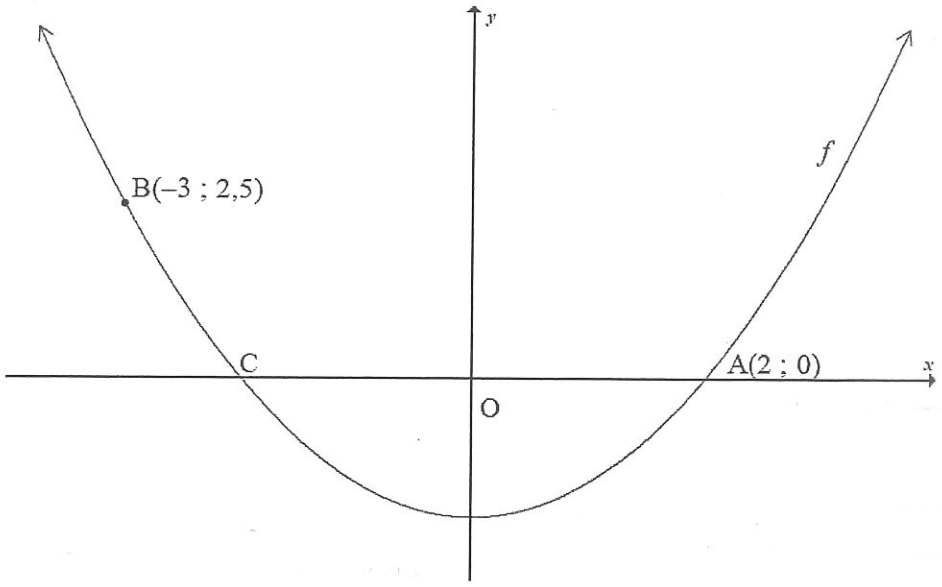
$x-int = 2$ ✓

$y-int = -4$ ✓

4.2 The graph of $f(x) = ax^2 + q$ is sketched below.

Points $A(2;0)$ and $B(-3;2,5)$ lie on the graph of f.

Points A and C are x-intercepts of f.



a) Write down the coordinates of C. (1)

$C(-2;0)$ ✓

b) Determine the equation of f. (4)

Sub $A(2;0)$ $0 = 4a + q$ $-(1)$ ✓

$B(-3;2,5)$ $\frac{5}{2} = 9a + q$ $-(2)$ ✓

$(2)-(1)$ $\frac{5}{2} = 5a$

$\frac{1}{2} = a$ ✓

$\therefore q = -2$

$f(x) = \frac{1}{2}x^2 - 2$ ✓

c) Write down the range of f. (1)

$y \geq -2$ ✓

d) Write down the range of h, where $h(x) = -f(x) - 2$. (1)

$y \leq 0$ CA

e) Determine the equation of an exponential function, $g(x) = b^x + q$, with range $y > -4$, that passes through point A and where $b > 0$. (3)

$q = -4$

Sub $(2;0)$: $0 = b^2 - 4$ ✓

$4 = b^2$

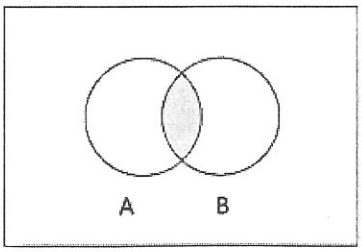
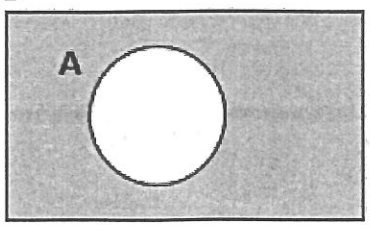
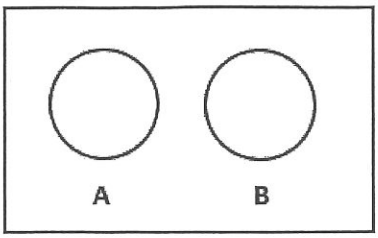
$2 = b$ ✓

$\therefore g(x) = 2^x - 4$ ✓

QUESTION 5 [14 Marks]

5.1 What word or expression from the list below, best describes the shaded area in the following Venn diagrams? (3)

- Intersection
- Union
- Complement
- Disjoint

Venn Diagram	Word/Expression
S 	intersection ✓
S 	complement ✓
S 	disjoint ✓

FORMULAE YOU MAY NEED:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

5.2 In a random chemistry experiment, R and S are two different events. It was found that $P(S) = \frac{4}{10}$, $P(not\ R) = \frac{3}{5}$, and $P(R\ or\ S) = \frac{4}{5}$.

a) Calculate $P(R)$. (1)

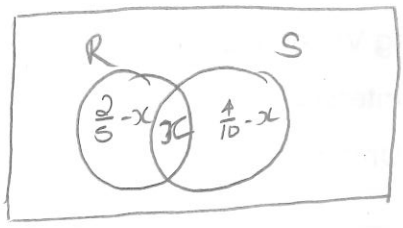
$P(R) = \frac{2}{5}$ ✓

b) Calculate $P(R\ and\ S)$. (3)

$\frac{2}{5} - x + x + \frac{4}{10} - x + \frac{1}{5} = 1$ ✓M

$1 - 1 = x$

$0 = x$ ✓ans



✓diag

OR

$P(R \cup S) = P(R) + P(S) - P(R \cap S)$

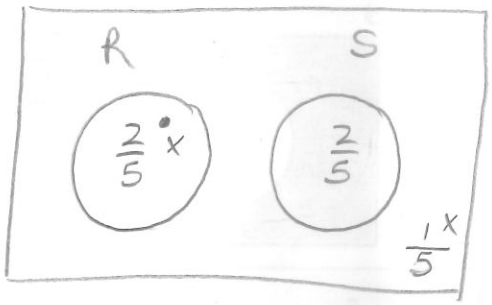
$\frac{4}{5} = \frac{2}{5} + \frac{4}{10} - P(R \cap S)$ ✓M

$P(R \cap S) = \frac{4}{5} - \frac{4}{5}$

$= 0$ ✓ ans

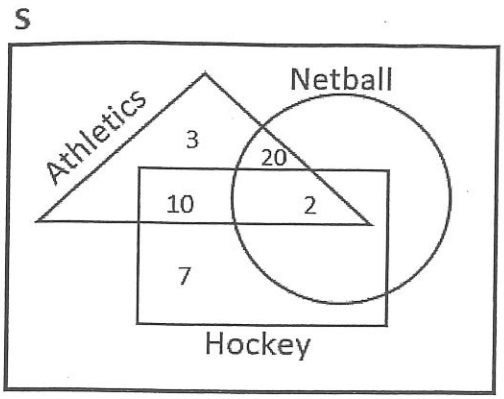
c) Hence, calculate $P(R\ and\ not\ S)$. (2)

$= \frac{2}{5}$ ✓ ans



1 mark if diag seen & correct

5.3 The Venn diagram below is a visual representation of a survey done with the high school learners at De La Salle Holy Cross College.



a) How many learners play hockey? (1)

19 ✓

b) How many learners do all three sports? (1)

2 ✓

c) If 50 learners took part in the survey, how many learners don't play any sport? (2)

$50 - (3 + 10 + 7 + 2 + 20)$ ✓ m

$= 8$ ✓ ans

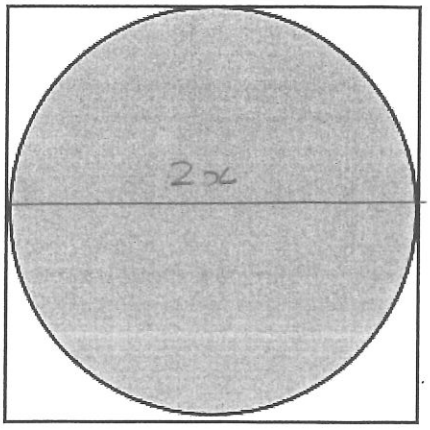
d) What is the probability that a learner selected at random will do athletics but not netball? (1)

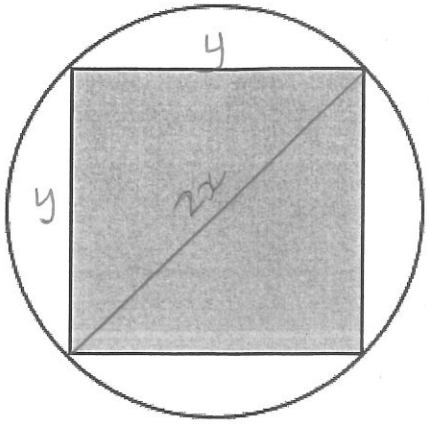
$= \frac{13}{50}$ ✓

QUESTION 6 [7 Marks]

Which is a better fit, a circle inscribed in a square or a square inscribed in a circle? (7)

The circles are congruent with a diameter of $2x$.





Area $\square = 4x^2$ ✓

Area $\circ = \pi x^2$ ✓

% shaded Area $= \frac{\pi x^2}{4x^2} \times 100$

$= 78,5\%$ ✓

OR

Space left $= 4x^2 - \pi x^2$

$= 0,86x$

∴ Circle inscribed in $\square$ is a better fit. ✓

Length $\square$: $2y^2 = 4x^2$

$y^2 = 2x^2$

$y = \sqrt{2}\,x$ ✓

∴ Area $\square = 2x^2$ ✓

Area $\circ = \pi x^2$

% shaded Area $= \frac{2x^2}{\pi x^2} \times 100$

$= 63,7\%$ ✓

OR

Space left $= \pi x^2 - 2x^2$

$= 1,14x^2$

TOTAL: [100]

According to Albert Einstein:

"It's not that I'm so smart, it's just that I stay with problems longer."